Input system:
Dependent variables:x_2, x_3, x_1
Independent variables:t_1, t_2
Equations:

$$\frac{\partial x_1}{\partial t_1} = (a_1 x_1^2 + b_1 x_2 + c_1)\sin(\mathrm{Hb}(x_1, x_2, x_3)) + (c_1 x_3^2 + b_1 x_1 + a_1)\cos(\mathrm{Hb}(x_1, x_2, x_3))$$

$$\frac{\partial x_1}{\partial t_2} = (a_2 x_1^3 + b_2 x_3 + c_2)\sin(\mathrm{Hb}(x_1, x_2, x_3)) + (c_2 x_3^2 + b_2 x_2 + a_2)\cos(\mathrm{Hb}(x_1, x_2, x_3))$$

$$\frac{\partial x_2}{\partial t_1} = (a_3 x_1^2 + b_3 x_2 + c_3)\sin(\mathrm{Hb}(x_1, x_2, x_3)) + (c_3 x_3^2 + b_3 x_1 + a_3)\cos(\mathrm{Hb}(x_1, x_2, x_3))$$

$$\frac{\partial x_2}{\partial t_2} = (a_4 x_1^3 + b_4 x_3 + c_4)\sin(\mathrm{Hb}(x_1, x_2, x_3)) + (c_4 x_3^2 + b_4 x_2 + a_4)\cos(\mathrm{Hb}(x_1, x_2, x_3))$$

$$\frac{\partial x_3}{\partial t_1} = (a_5 x_1^2 + b_5 x_2 + c_5)\sin(\mathrm{Hb}(x_1, x_2, x_3)) + (c_5 x_3^2 + b_5 x_1 + a_5)\cos(\mathrm{Hb}(x_1, x_2, x_3))$$

$$\frac{\partial x_3}{\partial t_2} = (a_6 x_1^3 + b_6 x_3 + c_6)\sin(\mathrm{Hb}(x_1, x_2, x_3)) + (c_6 x_3^2 + b_6 x_2 + a_6)\cos(\mathrm{Hb}(x_1, x_2, x_3))$$

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Result:
Dependent variables:$x_2, s_4, s_2, s_3, s_1, x_3, x_1$
Independent variables:t_1, t_2
Additional variables:

$$s_1 = \mathrm{Hb}(x_1, x_2, x_3)$$

$$s_2 = \mathrm{Hbi}(x_1, x_2, x_3)$$

$$s_3 = \sin(s_1)$$

$$s_4 = \cos(s_1)$$

Equations:

$$\frac{\partial x_1}{\partial t_1} = s_3 a_1 x_1^2 + s_4 c_1 x_3^2 + s_3 b_1 x_2 + s_4 b_1 x_1 + s_3 c_1 + s_4 a_1$$

$$\frac{\partial x_1}{\partial t_2} = s_3 a_2 x_1^3 + s_4 c_2 x_3^2 + s_3 b_2 x_3 + s_4 b_2 x_2 + s_3 c_2 + s_4 a_2$$

$$\frac{\partial x_2}{\partial t_1} = s_3 a_3 x_1^2 + s_4 c_3 x_3^2 + s_3 b_3 x_2 + s_4 b_3 x_1 + s_3 c_3 + s_4 a_3$$

$$\frac{\partial x_2}{\partial t_2} = s_3 a_4 x_1^3 + s_4 c_4 x_3^2 + s_3 b_4 x_3 + s_4 b_4 x_2 + s_3 c_4 + s_4 a_4$$

$$\frac{\partial x_3}{\partial t_1} = s_3 a_5 x_1^2 + s_4 c_5 x_3^2 + s_3 b_5 x_2 + s_4 b_5 x_1 + s_3 c_5 + s_4 a_5$$

$$\frac{\partial x_3}{\partial t_2} = s_3 a_6 x_1^3 + s_4 c_6 x_3^2 + s_3 b_6 x_3 + s_4 b_6 x_2 + s_3 c_6 + s_4 a_6$$

$$\begin{aligned}\frac{\partial s_1}{\partial t_1} = {} & (-1)s_1^3 s_2 s_3 a_5 x_1^2 + (-1)s_1^3 s_2 s_4 c_5 x_3^2 + (-1)s_1^2 s_2 s_3 a_3 x_1^2 + (-1)s_1^2 s_2 s_4 c_3 x_3^2 \\ & + (-1)s_1^3 s_2 s_3 b_5 x_2 + (-1)s_1^3 s_2 s_4 b_5 x_1 + (-1)s_1 s_2 s_3 a_1 x_1^2 \\ & + (-1)s_1 s_2 s_4 c_1 x_3^2 + (-1)s_1^2 s_2 s_3 b_3 x_2 + (-1)s_1^2 s_2 s_4 b_3 x_1 \\ & + (-1)s_1^3 s_2 s_3 c_5 + (-1)s_1^3 s_2 s_4 a_5 + (-1)s_1 s_2 s_3 b_1 x_2 + (-1)s_1 s_2 s_4 b_1 x_1 \\ & + (-1)s_1^2 s_2 s_3 c_3 + (-1)s_1^2 s_2 s_4 a_3 + (-1)s_1 s_2 s_3 c_1 + (-1)s_1 s_2 s_4 a_1\end{aligned}$$

$$\begin{aligned}\frac{\partial s_1}{\partial t_2} = {} & (-1)s_1^3 s_2 s_3 a_6 x_1^3 + (-1)s_1^2 s_2 s_3 a_4 x_1^3 + (-1)s_1^3 s_2 s_4 c_6 x_3^2 + (-1)s_1 s_2 s_3 a_2 x_1^3 \\ & + (-1)s_1^2 s_2 s_4 c_4 x_3^2 + (-1)s_1^3 s_2 s_3 b_6 x_3 + (-1)s_1^3 s_2 s_4 b_6 x_2 \\ & + (-1)s_1 s_2 s_4 c_2 x_3^2 + (-1)s_1^2 s_2 s_3 b_4 x_3 + (-1)s_1^2 s_2 s_4 b_4 x_2 \\ & + (-1)s_1^3 s_2 s_3 c_6 + (-1)s_1^3 s_2 s_4 a_6 + (-1)s_1 s_2 s_3 b_2 x_3 + (-1)s_1 s_2 s_4 b_2 x_2 \\ & + (-1)s_1^2 s_2 s_3 c_4 + (-1)s_1^2 s_2 s_4 a_4 + (-1)s_1 s_2 s_3 c_2 + (-1)s_1 s_2 s_4 a_2\end{aligned}$$

$$\begin{aligned}\frac{\partial s_2}{\partial t_1} = {} & 42 s_3 a_5 x_1^2 s_2^3 s_1^8 + 42 s_4 c_5 x_3^2 s_2^3 s_1^8 + 42 s_3 a_3 x_1^2 s_2^3 s_1^7 + 42 s_4 c_3 x_3^2 s_2^3 s_1^7 \\ & + 42 s_3 b_5 x_2 s_2^3 s_1^8 + 42 s_4 b_5 x_1 s_2^3 s_1^8 + 42 s_3 a_1 x_1^2 s_2^3 s_1^6 + 42 s_4 c_1 x_3^2 s_2^3 s_1^6 \\ & + 42 s_3 b_3 x_2 s_2^3 s_1^7 + 42 s_4 b_3 x_1 s_2^3 s_1^7 + 42 s_3 c_5 s_2^3 s_1^8 + 42 s_4 a_5 s_2^3 s_1^8 + 42 s_3 b_1 x_2 s_2^3 s_1^6 \\ & + 42 s_4 b_1 x_1 s_2^3 s_1^6 + 42 s_3 c_3 s_2^3 s_1^7 + 42 s_4 a_3 s_2^3 s_1^7 + 6 s_3 a_5 x_1^2 s_2^3 s_1^4 x_3 + 6 s_4 c_5 x_3^2 s_2^3 s_1^4 \\ & + 42 s_3 c_1 s_2^3 s_1^6 + 42 s_4 a_1 s_2^3 s_1^6 + 6 s_3 a_3 x_1^2 s_2^3 s_1^3 x_3 + 6 s_4 c_3 x_3^2 s_2^3 s_1^3 + 6 s_3 b_5 x_2 s_2^3 s_1^4 x_3 \\ & + 2 s_2^3 s_1^3 x_2 s_3 a_5 x_1^2 + 6 s_4 b_5 x_1 s_2^3 s_1^4 x_3 + 2 s_2^3 s_1^3 x_2 s_4 c_5 x_3^2 + 6 s_3 a_1 x_1^2 s_2^3 s_1^2 x_3 \\ & + 6 s_4 c_1 x_3^2 s_2^3 s_1^2 + 6 s_3 b_3 x_2 s_2^3 s_1^3 x_3 + 2 s_2^3 s_1^2 x_2 s_3 a_3 x_1^2 + 6 s_4 b_3 x_1 s_2^3 s_1^3 x_3 \\ & + 2 s_2^3 s_1^2 x_2 s_4 c_3 x_3^2 + 2 s_2^3 s_1^3 x_2^2 s_3 b_5 + 6 s_3 c_5 s_2^3 s_1^4 x_3 + 2 s_2^3 s_1^3 x_2 s_4 b_5 x_1 \\ & + 6 s_4 a_5 s_2^3 s_1^4 x_3 + 6 s_3 b_1 x_2 s_2^3 s_1^2 x_3 + 2 s_2^3 s_1 x_2 s_3 a_1 x_1^2 + 6 s_4 b_1 x_1 s_2^3 s_1^2 x_3 \\ & + 2 s_2^3 s_1 x_2 s_4 c_1 x_3^2 + 2 s_2^3 s_1^2 x_2^2 s_3 b_3 + 6 s_3 c_3 s_2^3 s_1^3 x_3 + 2 s_2^3 s_1^2 x_2 s_4 b_3 x_1 \\ & + 6 s_4 a_3 s_2^3 s_1^3 x_3 + 2 s_3 c_5 s_2^3 s_1^3 x_2 + 2 s_4 a_5 s_2^3 s_1^3 x_2 + 2 s_2^3 s_1 x_2^2 s_3 b_1 + 6 s_3 c_1 s_2^3 s_1^2 x_3 \\ & + 2 s_2^3 s_1 x_2 s_4 b_1 x_1 + 6 s_4 a_1 s_2^3 s_1^2 x_3 + 2 s_3 c_3 s_2^3 s_1^2 x_2 + 2 s_4 a_3 s_2^3 s_1^2 x_2 \\ & + (-2) s_1^2 s_2^2 s_3 a_5 x_1^2 + (-2) s_1^2 s_2^2 s_4 c_5 x_3^2 + 2 s_3 c_1 s_2^3 s_1 x_2 + 2 s_4 a_1 s_2^3 s_1 x_2 \\ & + (-1) s_1 s_2^2 s_3 a_3 x_1^2 + (-1) s_1 s_2^2 s_4 c_3 x_3^2 + (-2) s_1^2 s_2^2 s_3 b_5 x_2 + (-2) s_1^2 s_2^2 s_4 b_5 x_1 \\ & + (-1) s_2^2 s_3 a_1 x_1^2 + (-1) s_2^2 s_4 c_1 x_3^2 + (-1) s_1 s_2^2 s_3 b_3 x_2 + (-1) s_1 s_2^2 s_4 b_3 x_1 \\ & + (-2) s_1^2 s_2^2 s_3 c_5 + (-2) s_1^2 s_2^2 s_4 a_5 + (-1) s_2^2 s_3 b_1 x_2 + (-1) s_2^2 s_4 b_1 x_1 \\ & + (-1) s_1 s_2^2 s_3 c_3 + (-1) s_1 s_2^2 s_4 a_3 + (-1) s_2^2 s_3 c_1 + (-1) s_2^2 s_4 a_1\end{aligned}$$

$$\frac{\partial s_2}{\partial t_2} = 42s_3a_6x_1^3s_2^3s_1^8 + 42s_3a_4x_1^3s_2^3s_1^7 + 42s_4c_6x_3^2s_2^3s_1^8 + 42s_3a_2x_1^3s_2^3s_1^6$$
$$+ 42s_4c_4x_3^2s_2^3s_1^7 + 42s_3b_6x_3s_2^3s_1^8 + 42s_4b_6x_2s_2^3s_1^8 + 42s_4c_2x_3^2s_2^3s_1^6$$
$$+ 42s_3b_4x_3s_2^3s_1^7 + 42s_4b_4x_2s_2^3s_1^7 + 6s_3a_6x_1^3s_2^3s_1^4x_3 + 42s_3c_6s_2^3s_1^8$$
$$+42s_4a_6s_2^3s_1^8+42s_3b_2x_3s_2^3s_1^6+42s_4b_2x_2s_2^3s_1^6+6s_3a_4x_1^3s_2^3s_1^3x_3+42s_3c_4s_2^3s_1^7$$
$$+ 42s_4a_4s_2^3s_1^7 + 2s_2^3s_1^3x_2s_3a_6x_1^3 + 6s_4c_6x_3^3s_2^3s_1^4 + 6s_3a_2x_1^3s_2^3s_1^2x_3$$
$$+ 42s_3c_2s_2^3s_1^6 + 42s_4a_2s_2^3s_1^6 + 2s_2^3s_1^2x_2s_3a_4x_1^3 + 6s_4c_4x_3^3s_2^3s_1^3 + 6s_3b_6x_3^2s_2^3s_1^4$$
$$+ 6s_4b_6x_2s_2^3s_1^4x_3 + 2s_2^3s_1^3x_2s_4c_6x_3^2 + 2s_2^3s_1x_2s_3a_2x_1^3 + 6s_4c_2x_3^3s_2^3s_1^2$$
$$+ 6s_3b_4x_3^2s_2^3s_1^3 + 6s_4b_4x_2s_2^3s_1^3x_3 + 2s_2^3s_1^2x_2s_4c_4x_3^2 + 2s_2^3s_1^3x_2s_3b_6x_3$$
$$+6s_3c_6s_2^3s_1^4x_3+2s_2^3s_1^3x_2^2s_4b_6+6s_4a_6s_2^3s_1^4x_3+6s_3b_2x_3^2s_2^3s_1^2+6s_4b_2x_2s_2^3s_1^2x_3$$
$$+ 2s_2^3s_1x_2s_4c_2x_3^2 + 2s_2^3s_1^2x_2s_3b_4x_3 + 6s_3c_4s_2^3s_1^3x_3 + 2s_2^3s_1^2x_2^2s_4b_4$$
$$+ 6s_4a_4s_2^3s_1^3x_3 + 2s_3c_6s_2^3s_1^3x_2 + (-2)s_1^2s_2^2s_3a_6x_1^3 + 2s_4a_6s_2^3s_1^3x_2$$
$$+ 2s_2^3s_1x_2s_3b_2x_3 + 6s_3c_2s_2^3s_1^2x_3 + 2s_2^3s_1x_2^2s_4b_2 + 6s_4a_2s_2^3s_1^2x_3$$
$$+ 2s_3c_4s_2^3s_1^2x_2 + (-1)s_1s_2^2s_3a_4x_1^3 + 2s_4a_4s_2^3s_1^2x_2 + (-2)s_1^2s_2^2s_4c_6x_3^2$$
$$+ 2s_3c_2s_2^3s_1x_2 + (-1)s_2^2s_3a_2x_1^3 + 2s_4a_2s_2^3s_1x_2 + (-1)s_1s_2^2s_4c_4x_3^2$$
$$+ (-2)s_1^2s_2^2s_3b_6x_3 + (-2)s_1^2s_2^2s_4b_6x_2 + (-1)s_2^2s_4c_2x_3^2 + (-1)s_1s_2^2s_3b_4x_3$$
$$+ (-1)s_1s_2^2s_4b_4x_2 + (-2)s_1^2s_2^2s_3c_6 + (-2)s_1^2s_2^2s_4a_6 + (-1)s_2^2s_3b_2x_3$$
$$+(-1)s_2^2s_4b_2x_2+(-1)s_1s_2^2s_3c_4+(-1)s_1s_2^2s_4a_4+(-1)s_2^2s_3c_2+(-1)s_2^2s_4a_2$$

$$\frac{\partial s_3}{\partial t_1} = (-1)s_4s_1^3s_2s_3a_5x_1^2+(-1)s_4^2s_1^3s_2c_5x_3^2+(-1)s_4s_1^2s_2s_3a_3x_1^2+(-1)s_4^2s_1^2s_2c_3x_3^2$$
$$+ (-1)s_4s_1^3s_2s_3b_5x_2 + (-1)s_4^2s_1^3s_2b_5x_1 + (-1)s_4s_1s_2s_3a_1x_1^2$$
$$+ (-1)s_4^2s_1s_2c_1x_3^2 + (-1)s_4s_1^2s_2s_3b_3x_2 + (-1)s_4^2s_1^2s_2b_3x_1$$
$$+ (-1)s_4s_1^3s_2s_3c_5 + (-1)s_4^2s_1^3s_2a_5 + (-1)s_4s_1s_2s_3b_1x_2 + (-1)s_4^2s_1s_2b_1x_1$$
$$+ (-1)s_4s_1^2s_2s_3c_3 + (-1)s_4^2s_1^2s_2a_3 + (-1)s_4s_1s_2s_3c_1 + (-1)s_4^2s_1s_2a_1$$

$$\frac{\partial s_3}{\partial t_2} = (-1)s_4s_1^3s_2s_3a_6x_1^3 + (-1)s_4s_1^2s_2s_3a_4x_1^3 + (-1)s_4^2s_1^3s_2c_6x_3^2$$
$$+ (-1)s_4s_1s_2s_3a_2x_1^3 + (-1)s_4^2s_1^2s_2c_4x_3^2 + (-1)s_4s_1^3s_2s_3b_6x_3$$
$$+(-1)s_4^2s_1^3s_2b_6x_2+(-1)s_4^2s_1s_2c_2x_3^2+(-1)s_4s_1^2s_2s_3b_4x_3+(-1)s_4^2s_1^2s_2b_4x_2$$
$$+ (-1)s_4s_1^3s_2s_3c_6 + (-1)s_4^2s_1^3s_2a_6 + (-1)s_4s_1s_2s_3b_2x_3 + (-1)s_4^2s_1s_2b_2x_2$$
$$+ (-1)s_4s_1^2s_2s_3c_4 + (-1)s_4^2s_1^2s_2a_4 + (-1)s_4s_1s_2s_3c_2 + (-1)s_4^2s_1s_2a_2$$

$$\frac{\partial s_4}{\partial t_1} = s_3^2s_1^3s_2a_5x_1^2 + s_3s_1^3s_2s_4c_5x_3^2 + s_3^2s_1^2s_2a_3x_1^2 + s_3s_1^2s_2s_4c_3x_3^2$$
$$+ s_3^2s_1^3s_2b_5x_2 + s_3s_1^3s_2s_4b_5x_1 + s_3^2s_1s_2a_1x_1^2 + s_3s_1s_2s_4c_1x_3^2$$
$$+ s_3^2s_1^2s_2b_3x_2 + s_3s_1^2s_2s_4b_3x_1 + s_3^2s_1^3s_2c_5 + s_3s_1^3s_2s_4a_5 + s_3^2s_1s_2b_1x_2$$
$$+ s_3s_1s_2s_4b_1x_1 + s_3^2s_1^2s_2c_3 + s_3s_1^2s_2s_4a_3 + s_3^2s_1s_2c_1 + s_3s_1s_2s_4a_1$$

$$\frac{\partial s_4}{\partial t_2} = s_3^2s_1^3s_2a_6x_1^3 + s_3^2s_1^2s_2a_4x_1^3 + s_3s_1^3s_2s_4c_6x_3^2 + s_3^2s_1s_2a_2x_1^3$$
$$+ s_3s_1^2s_2s_4c_4x_3^2 + s_3^2s_1^3s_2b_6x_3 + s_3s_1^3s_2s_4b_6x_2 + s_3s_1s_2s_4c_2x_3^2$$
$$+ s_3^2s_1^2s_2b_4x_3 + s_3s_1^2s_2s_4b_4x_2 + s_3^2s_1^3s_2c_6 + s_3s_1^3s_2s_4a_6 + s_3^2s_1s_2b_2x_3$$
$$+ s_3s_1s_2s_4b_2x_2 + s_3^2s_1^2s_2c_4 + s_3s_1^2s_2s_4a_4 + s_3^2s_1s_2c_2 + s_3s_1s_2s_4a_2$$